

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical
Practical – XII

Roll No.: T009	Name: Siddhesh Kamble
Class: TYIT	Batch: 01
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Practical 12: Analysis of Variance and Experimental Design

Part A: Analysis of Variance (ANOVA)

1. Load the Retail and warehouse Sale.csv dataset using Pandas.
2. Display the first five records and identify the number of rows and columns.
3. Display the column names and data types.
4. Check for missing values and duplicate records.
5. Display the different categories present in the ITEM TYPE column.
6. Calculate the average RETAIL SALES for each ITEM TYPE.
7. Create a box plot to compare RETAIL SALES across different ITEM TYPE categories.
8. Formulate the null and alternative hypotheses to determine whether the average retail sales differ among item types.
9. Perform one-way ANOVA using ITEM TYPE as the factor and RETAIL SALES as the response variable.
10. Calculate and display the F-statistic and p-value.
11. Compare the p-value with the significance level of 0.05.
12. State whether the null hypothesis is rejected or not rejected.
13. Write the conclusion about the difference in retail sales among different item types.

Code :

```
# Practical 12: Analysis of Variance (ANOVA)
# Part A: One-Way ANOVA
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import f_oneway
file_name = "Retail and warehouse Sale.csv"
df = pd.read_csv(file_name)
print("\nFIRST FIVE RECORDS")
print("-----")
print(df.head())
print("\nNumber of Rows      :", df.shape[0])
print("Number of Columns  :", df.shape[1])
print("\nCOLUMN NAMES")
print("-----")
print(df.columns.tolist())
print("\nDATA TYPES")
print("-----")
print(df.dtypes)
print("\nMISSING VALUES")
print("-----")
print(df.isnull().sum())
print("\nDUPLICATE RECORDS")
print("-----")
print(df.duplicated().sum())
df["RETAIL SALES"] = df["RETAIL SALES"].fillna(
    df["RETAIL SALES"].median()
)
print("\nITEM TYPE CATEGORIES")
print("-----")
print(df["ITEM TYPE"].unique())
print("\nAVERAGE RETAIL SALES BY ITEM TYPE")
print("-----")
average_sales = (
    df.groupby("ITEM TYPE")["RETAIL SALES"]
    .mean()
    .sort_values(ascending=False)
)
print(average_sales)
plt.figure(figsize=(10, 6))
```

```

sns.boxplot(
    data=df,
    x="ITEM TYPE",
    y="RETAIL SALES"
)
plt.title("Retail Sales by Item Type")
plt.xlabel("Item Type")
plt.ylabel("Retail Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
print("\nHYPOTHESES")
print("-----")
print("Null Hypothesis (H0):")
print(
    "The average retail sales are equal "
    "among all ITEM TYPE categories."
)
print("\nAlternative Hypothesis (H1):")
print(
    "At least one ITEM TYPE has a different "
    "average retail sales."
)
groups = []
for item_type in df["ITEM TYPE"].unique():
    group = df[
        df["ITEM TYPE"] == item_type
    ]["RETAIL SALES"]
    groups.append(group)
f_statistic, p_value = f_oneway(*groups)
print("\nONE-WAY ANOVA RESULTS")
print("-----")
print("F-Statistic :", f_statistic)
print("P-Value      :", p_value)
alpha = 0.05
print("\nSIGNIFICANCE TEST")
print("-----")
print("Significance Level :", alpha)
if p_value < alpha:
    print("P-value < 0.05")
    print(
        "Decision: Reject the Null Hypothesis."
    )
else:
    print("P-value >= 0.05")
    print(
        "Decision: Do Not Reject the Null Hypothesis."
    )
print("\nCONCLUSION")
print("-----")
if p_value < alpha:
    print(
        "There is a statistically significant difference "
        "in average retail sales among the different "
        "ITEM TYPE categories."
    )
else:
    print(
        "There is no statistically significant difference "
        "in average retail sales among the different "
        "ITEM TYPE categories."
    )

```

Output :

FIRST FIVE RECORDS

	YEAR	MONTH	SUPPLIER	ITEM CODE	\
0	2020	1	REPUBLIC NATIONAL DISTRIBUTING CO	100009	
1	2020	1	PWSWN INC	100024	
2	2020	1	RELIABLE CHURCHILL LLLP	1001	
3	2020	1	LANTERNA DISTRIBUTORS INC	100145	
4	2020	1	DIONYSOS IMPORTS INC	100293	

	ITEM DESCRIPTION	ITEM TYPE	RETAIL SALES	\
0	BOOTLEG RED - 750ML	WINE	0.00	
1	MOMENT DE PLAISIR - 750ML	WINE	0.00	
2	S SMITH ORGANIC PEAR CIDER - 18.7OZ	BEER	0.00	
3	SCHLINK HAUS KABINETT - 750ML	WINE	0.00	
4	SANTORINI GAVALA WHITE - 750ML	WINE	0.82	

	RETAIL TRANSFERS	WAREHOUSE SALES
0	0.0	2.0
1	1.0	4.0
2	0.0	1.0
3	0.0	1.0
4	0.0	0.0

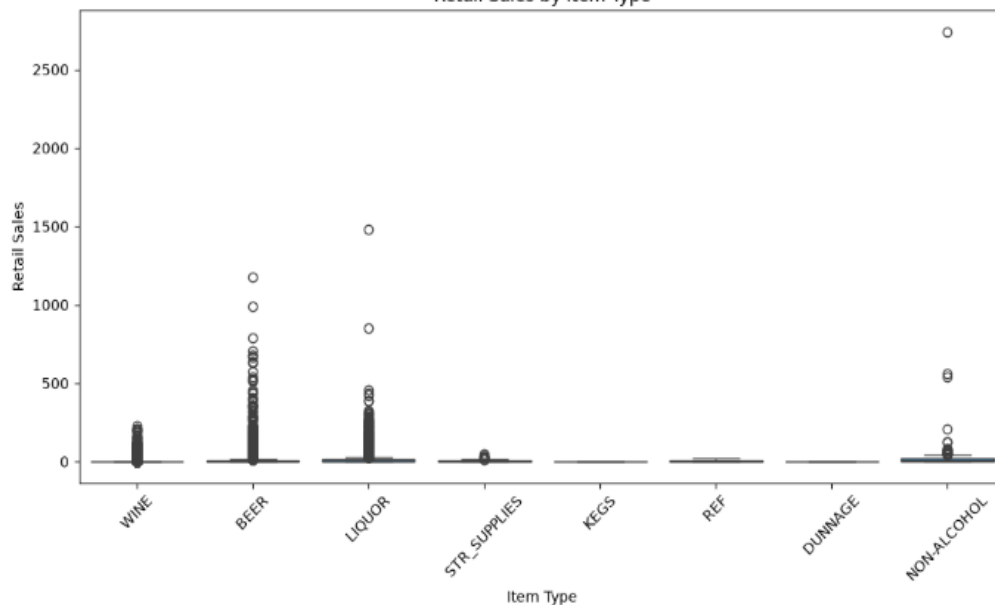
Number of Rows : 30000

WINE 3.195334
 BEER 0.000000
 LIQUOR 0.000000
 STR_SUPPLIES 0.000000
 KECS 0.000000
 REF 0.000000
 DUNNAGE 0.000000
 NON-ALCOHOL 0.000000

Name: RETAIL SALES, dtype: float64

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output settings...

Retail Sales by Item Type



HYPOTHESES

Null Hypothesis (H0):

The average retail sales are equal among all ITEM TYPE categories.

Alternative Hypothesis (H1):

At least one ITEM TYPE has a different average retail sales.

ONE-WAY ANOVA RESULTS

F-Statistic : 123.39469052328639

P-Value : 1.3337307137369178e-179

SIGNIFICANCE TEST

Significance Level : 0.05

P-value < 0.05

Decision: Reject the Null Hypothesis.

CONCLUSION

There is a statistically significant difference in average retail sales among the different ITEM TYPE categories.

Part B: Factorial Experiment

14. Select ITEM TYPE and YEAR as two factors and RETAIL SALES as the response variable.
15. Display the average retail sales for each combination of ITEM TYPE and YEAR.
16. Create a suitable visualization to compare retail sales for different item types across years.
17. Analyze the main effect of ITEM TYPE on retail sales.
18. Analyze the main effect of YEAR on retail sales.
19. Analyze the interaction effect between ITEM TYPE and YEAR.
20. Perform a suitable factorial ANOVA using the selected factors.
21. Display the statistical results and p-values for the main and interaction effects.
22. Identify whether ITEM TYPE, YEAR, or their interaction has a significant effect on RETAIL SALES.
23. Write the final business conclusion based on the ANOVA and factorial experiment results.

Code:

```
# Practical 12: Factorial Experiment
# Part B: Factorial ANOVA
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from statsmodels.formula.api import ols
file_name = "Retail and warehouse Sale.csv"
df = pd.read_csv(file_name)
df["RETAIL SALES"] = df["RETAIL SALES"].fillna(
    df["RETAIL SALES"].median()
)
print("\nFACTORIAL EXPERIMENT")
print("=====")
print("\nFactor 1 : ITEM TYPE")
print("Factor 2 : YEAR")
print("Response : RETAIL SALES")
print("\nYEAR VALUES")
print("-----")
print(df["YEAR"].unique())
print(
    "\nNumber of different years:",
    df["YEAR"].nunique()
)
print("\nAVERAGE RETAIL SALES")
print("ITEM TYPE AND YEAR")
print("=====")
average_sales = (
    df.groupby(
        ["ITEM TYPE", "YEAR"]
    )["RETAIL SALES"]
    .mean()
    .reset_index()
)
print(average_sales)
plt.figure(figsize=(12, 6))
sns.barplot(
    data=df,
    x="ITEM TYPE",
    y="RETAIL SALES",
    hue="YEAR",
    errorbar=None
)
plt.title(
    "Retail Sales by Item Type and Year"
)
plt.xlabel("Item Type")
plt.ylabel("Average Retail Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
print("\nMAIN EFFECT OF ITEM TYPE")
print("=====")
item_type_effect = (
    df.groupby("ITEM TYPE")["RETAIL SALES"]
```

```

        .mean()
    )
    print(item_type_effect)
    print("\nMAIN EFFECT OF YEAR")
    print("=====")
    year_effect = (
        df.groupby("YEAR")["RETAIL SALES"]
        .mean()
    )
    print(year_effect)
    print("\nFACTORIAL ANOVA")
    print("=====")
    if df["YEAR"].nunique() < 2:
        print("\nWARNING")
        print("-----")
        print(
            "The YEAR column contains only one unique value:"
        )
        print(df["YEAR"].unique())
        print(
            "\nTherefore, a two-factor ANOVA cannot be performed."
        )
        print("\nReason:")
        print(
            "YEAR must contain at least two different levels "
            "to calculate its main effect and interaction."
        )
        print("\nCannot statistically calculate:")
        print("1. Main effect of YEAR")
        print("2. ITEM TYPE × YEAR interaction")

        print(
            "\nThe ITEM TYPE effect has already been tested "
            "using one-way ANOVA in Part A."
        )
    else:
        model = ols(
            "Q('RETAIL SALES') ~ "
            "C(Q('ITEM TYPE')) * C(Q('YEAR'))",
            data=df
        ).fit()
        anova_table = sm.stats.anova_lm(
            model,
            typ=2
        )
        print("\nANOVA TABLE")
        print("-----")
        print(anova_table)
        print("\nP-VALUES")
        print("-----")
        print(anova_table["PR(>F)"])
        print("\nSIGNIFICANT EFFECTS")
        print("-----")
        for effect, p_value in anova_table["PR(>F)"].items():
            if pd.notna(p_value):
                if p_value < 0.05:
                    print(
                        effect,
                        ": Significant"
                    )
                else:
                    print(
                        effect,
                        ": Not Significant"
                    )
        print("\nFINAL BUSINESS CONCLUSION")
        print("=====")
        if df["YEAR"].nunique() < 2:
            print(
                "The uploaded dataset contains only one year, 2020."
            )
            print(
                "Therefore, a complete factorial experiment using "
                "ITEM TYPE and YEAR cannot be performed."
            )

```

```

    )
    print(
        "A dataset containing at least two different years "
        "is required to test the YEAR effect and the "
        "ITEM TYPE × YEAR interaction."
    )
else:
    print(
        "The factorial ANOVA results can be used to determine "
        "the significant effects of ITEM TYPE, YEAR, and "
        "their interaction."
    )

```

Output :

```

---
FACTORIAL EXPERIMENT
=====

Factor 1 : ITEM TYPE
Factor 2 : YEAR
Response : RETAIL SALES

YEAR VALUES
-----
[2020]

Number of different years: 1

AVERAGE RETAIL SALES
ITEM TYPE AND YEAR
=====

```

	ITEM TYPE	YEAR	RETAIL SALES
0	BEER	2020	14.118748
1	DUNNAGE	2020	0.000000
2	KEGS	2020	0.000000
3	LIQUOR	2020	13.635171
4	NON-ALCOHOL	2020	31.596204
5	REF	2020	5.783750
6	STR_SUPPLIES	2020	5.485714
7	WINE	2020	3.195334

```

---

---

```

Item Type	Average Retail Sales
WINE	3.195334
BEER	14.118748
LIQUOR	13.635171
STR_SUPPLIES	5.485714
KEGS	0.000000
REF	5.783750
DUNNAGE	0.000000
NON-ALCOHOL	31.596204

```

---
MAIN EFFECT OF ITEM TYPE
=====
ITEM TYPE
BEER          14.118748
DUNNAGE       0.000000
KEGS          0.000000
LIQUOR        13.635171
NON-ALCOHOL   31.596204
REF           5.783750
STR_SUPPLIES  5.485714
WINE          3.195334
Name: RETAIL SALES, dtype: float64

MAIN EFFECT OF YEAR
=====
YEAR
2020    6.93957
Name: RETAIL SALES, dtype: float64

FACTORIAL ANOVA
=====

WARNING
-----
...

The uploaded dataset contains only one year, 2020.
Therefore, a complete factorial experiment using ITEM TYPE and YEAR cannot be performed.
A dataset containing at least two different years is required to test the YEAR effect and the ITEM TYPE × YEAR interaction.
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```